

WILSON LIU

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EDUCATION

Carnegie Mellon University

M.S. in Electrical and Computer Engineering

Pittsburgh, PA

Dec 2023 – May 2025

- **Relevant Coursework:** Computer Networks, Data Analytics, Neural Signal Processing, Big Data Science, Information Security

Rensselaer Polytechnic Institute

B.S. in Computer Science & Mathematics

Troy, NY

Sep 2019 – Dec 2023

- **Relevant Coursework:** Operating Systems, Machine Learning, Numerical Computing, Database Systems

PROFESSIONAL EXPERIENCE

Innovation AI

Backend/Full Stack Software Engineer Intern

Remote, PA

Apr 2025 – May 2025

- Enabled dynamic video streaming by designing RESTful APIs in **Spring Boot** and integrating file system logic, improving backend media access speed by 40%.
- Delivered seamless frontend experience by developing **Vue.js** components with Vuetify for interactive video playback and selection.
- Boosted debugging efficiency with real-time API usage logs, reducing incident resolution time by 25%.

Peppr Inc.

Product Intern

Los Angeles, CA

May 2024 – Aug 2024

- Accelerated feature rollout by 30% by prioritizing development based on user interviews and competitive research across 5 major POS platforms.
- Increased user satisfaction through UI/UX redesign of the POS dashboard and backend portal, resolving 80% of design pain points reported in feedback.
- Ensured 100% uptime during onboarding by managing on-site setup of networks, printers, and training for 10+ restaurant teams.
- Delivered actionable insights to product leadership by conducting cross-functional analysis of sales trends, usage behavior, and operational gaps.

Meituan Inc.

Data Analyst Intern

Shanghai, China

Feb 2022 – Jul 2022

- Reduced salary calculation turnaround by 60% by building automated performance evaluation tools using **SQL** and **Excel**.
- Delivered monthly business intelligence reports using **MySQL** and **Tableau**, supporting 20+ stakeholders in marketing and strategy.
- Analyzed 50+ competitors' financing and operations to support executive-level investment and expansion decisions.

PROJECTS EXPERIENCE

Matrix Transposition Optimization

Mar 2025 – May 2025

- Achieved 5× speedup for dense matrices using loop blocking + **OpenMP**, reaching 15 GB/s throughput on DDR4 memory systems.
- Tuned CSR format transposition via **Intel MKL**, ensuring efficient handling of large sparse matrices in ML pipelines.

AI-Based Badminton Performance Analytics

Jan 2025 – May 2025

- Delivered real-time shot tracking and spatial analytics using **YOLOv8** and **ResNet34**; improved trajectory smoothing under occlusion by 35%.
- Enabled coaches to visualize player heatmaps and footwork patterns with a custom OpenCV dashboard.

GPU-Accelerated Canny Edge Detection

Jan 2024 – May 2024

- Achieved 81% speedup vs. CPU baseline through **CUDA** parallelism and kernel fusion for high-res image processing (4032×3024).
- Minimized data latency by retaining intermediate results in GPU memory, reducing per-image transfer overhead by 138 MB.

SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL, MATLAB, LaTeX, Maple
- **Frameworks & Tools:** Spring Boot, Vue.js, CUDA, YOLO, ResNet, OpenCV, MySQL, Tableau, Intel MKL, OpenMP